House Prices Prediction: A Statistical Journey from Inference to Neural Networks

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Abstract

This report studies the prediction of residential house prices using the Kaggle *House Prices: Advanced Regression Techniques* dataset. Instead of directly treating the task as a black-box prediction problem, the analysis follows a progressive statistical path. We begin by understanding the distribution of house prices, then apply classical inference, ANOVA, factorial design, parametric regression, and finally a neural network model for Kaggle prediction. The goal is not only to obtain predictions, but also to understand which housing characteristics matter, how they interact, and how statistical and machine learning methods complement each other.

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1 Introduction

Predicting house prices is a rich statistical problem because the price of a home is rarely determined by a single factor. A house is a combination of physical size, construction quality, comfort, location-related characteristics, and many smaller details that together shape its market value. In this project, the objective is to predict the sale price of houses using statistical and machine learning methods, while also interpreting the role of the most relevant explanatory variables.

The dataset used in this work contains observations from the Ames housing market. Each house is described by numerical features, such as living area and basement size, and categorical or ordinal features, such as overall quality, kitchen quality, exterior quality, and fireplace quality. The target variable is `SalePrice`, the final sale price of each house.

The analysis follows the structure of the course project. First, classical statistical inference is used to understand the distribution of the target variable and selected explanatory variables. Then, ANOVA is used to identify statistically significant features. A 2^k factorial design is applied to study selected binary factors. After that, a parametric regression model is built using significant ANOVA features and selected numerical variables. Finally, a neural network model is trained using all available features to generate the final Kaggle submission file.

A key idea throughout the project is that prediction and interpretation should support each other. The statistical steps help us understand the data before moving toward more flexible machine learning models.

2 Data Overview and Target Transformation

The analysis begins by loading the training and test datasets. The training set contains 1460 observations and 81 columns, while the test set contains 1459 observations and 80 columns. The additional column in the training set is the target variable, `SalePrice`.

Before building any model, it is important to understand the shape of the target variable. House prices are positive monetary values, and such variables often show a right-skewed distribution: many houses have moderate prices, while a smaller number of houses are very expensive.

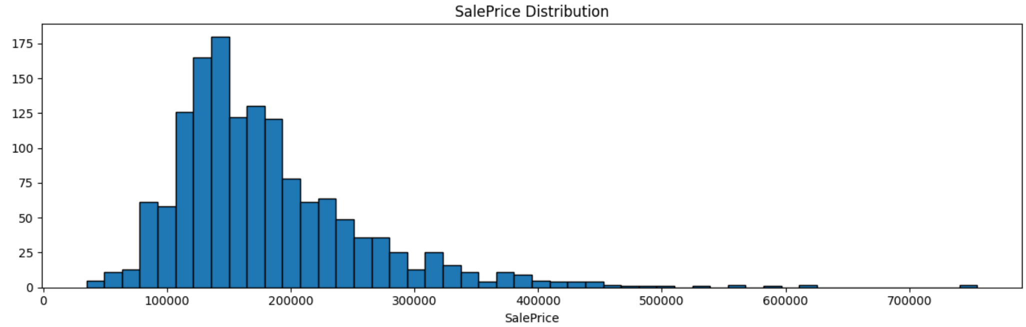

Figure 1: Distribution of the original SalePrice variable. The distribution is strongly right-skewed, showing that a few houses have very high sale prices.

The original `SalePrice` distribution was heavily right-skewed, with a skewness of approximately 1.883. This means that the data had a long right tail caused by a small number of very expensive houses. Such skewness can make statistical inference and linear modeling less stable, because many classical methods work better when the response variable is closer to a normal distribution.

To address this, a logarithmic transformation was applied:

$$\texttt{LogSalePrice} = \log(1 + \texttt{SalePrice}). \tag{1}$$

The use of `log1p` is appropriate because it is numerically stable and corresponds to the inverse transformation `expm1`, which will later be used to convert predicted log-prices back to sale prices.

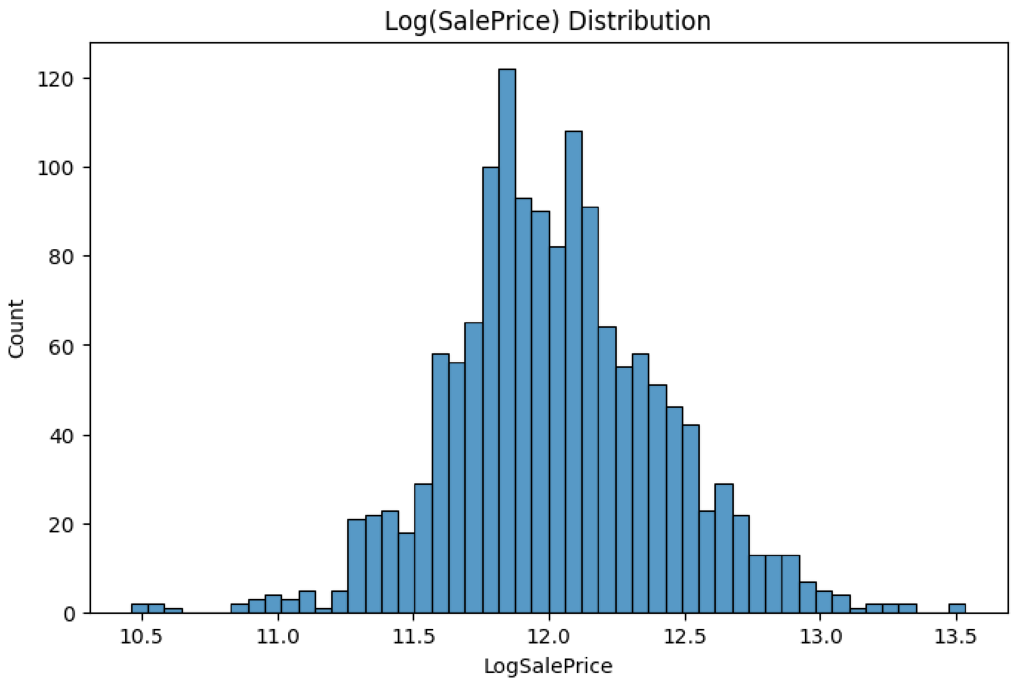

Figure 2: Distribution of LogSalePrice after logarithmic transformation. The transformed target is much closer to a bell-shaped distribution.

After transformation, the skewness decreased to approximately 0.121. This is a major improvement: the transformed price distribution is much more symmetric and closer to normal. Therefore, `LogSalePrice` is used as the main response variable in the statistical analysis and modeling steps.

3 Classical Statistical Inference

The first statistical step is to summarize the transformed target variable. This provides a baseline understanding of the central tendency and variability of house prices after the logarithmic transformation.

3.1 Sample Mean and Variance of LogSalePrice

The sample mean of `LogSalePrice` was computed as:

$$\bar{x} \approx 12.0241. \tag{2}$$

The sample variance was:

$$s^2 \approx 0.1596. \tag{3}$$

These two values summarize the transformed target variable. The mean represents the average log-transformed sale price, while the variance measures how dispersed the log-prices are around this average.

Table 1: Basic statistics for LogSalePrice

Statistic	Value
Mean	12.0241
Variance	0.1596
Skewness after log transformation	0.1213

From a storytelling perspective, this step is important because it marks the transition from raw house prices, which are difficult to model directly, to a more statistically manageable response variable. The log transformation does not remove the meaning of price; rather, it allows the analysis to focus on relative differences in price in a more stable way.

3.2 Confidence Interval for the Mean LogSalePrice

To estimate the range in which the true population mean of `LogSalePrice` is likely to lie, a 95% confidence interval was computed. Since the sample size is large, a normal approximation was used:

$$\bar{x} \pm z_{0.975} \frac{s}{\sqrt{n}}. \tag{4}$$

The resulting 95% confidence interval was approximately:

$$[12.0036,\ 12.0445]. \tag{5}$$

Table 2: 95% confidence interval for the mean of LogSalePrice

Quantity	Value
Lower bound	12.0036
Upper bound	12.0445
Confidence level	95%

This interval is relatively narrow, which is expected because the dataset contains 1460 training observations. It suggests that the average log-transformed house price is estimated with good precision.

3.3 Hypothesis Testing on the Mean LogSalePrice

Next, a one-sample t-test was performed to test whether the mean of `LogSalePrice` is significantly different from the reference value 12. The hypotheses were:

$$H_0 : \mu = 12, \tag{6}$$
$$H_1 : \mu \neq 12. \tag{7}$$

This is a two-sided test because the goal is to check whether the mean is different from 12 in either direction. The test produced:

Table 3: One-sample hypothesis test for the mean of LogSalePrice

Quantity	Value
Test statistic	2.3012
p-value	0.0215
Significance level	0.05
Decision	Reject H_0

Since the p-value is approximately 0.0215, which is below the significance level of 0.05, the null hypothesis is rejected. This means that there is sufficient statistical evidence to conclude that the mean of `LogSalePrice` is significantly different from 12.

This result is statistically significant, but its interpretation should remain practical. The estimated mean is close to 12, but the large sample size allows even a relatively small difference to be detected. Therefore, the result indicates that 12 is not an exact representation of the average log-transformed sale price in this dataset.

3.4 Exploring an Explanatory Variable: GrLivArea

After studying the target variable, the analysis also considered `GrLivArea`, which represents the above-ground living area of a house. This feature is important because living area is intuitively linked to price: larger houses are usually more expensive.

The computed statistics were:

Table 4: Basic statistics for GrLivArea

Statistic	Value
Mean	1515.46
Variance	276129.63

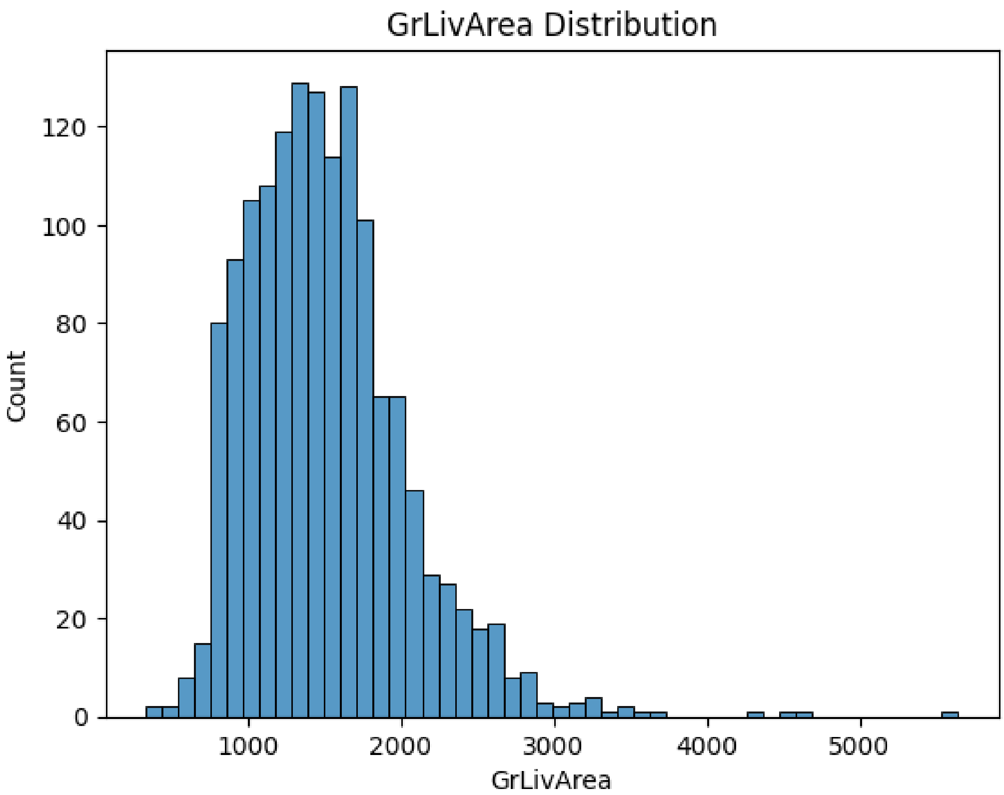

Figure 3: Distribution of GrLivArea. Most houses have a moderate living area, while a few houses have much larger above-ground living areas.

The distribution of `GrLivArea` is also right-skewed. Most houses have moderate living areas, while a smaller number of properties have very large living spaces. This pattern resembles the original price distribution and supports the intuition that large houses may contribute to higher sale prices.

3.5 Summary of Part 1

The first part of the analysis established a clean statistical foundation for the rest of the project. The original `SalePrice` variable was strongly right-skewed, so a logarithmic transformation was applied. The transformed variable, `LogSalePrice`, became much closer to normal, making it more appropriate for inference and modeling.

The confidence interval showed that the mean log-price is estimated precisely, and the hypothesis test indicated that the average `LogSalePrice` is significantly different from the reference value 12. Finally, the analysis of `GrLivArea` introduced the idea that explanatory variables may also have skewed distributions and that physical size is likely to be an important factor in price prediction.

This first step prepares the ground for the next stage: using ANOVA to identify which housing characteristics have statistically significant effects on `LogSalePrice`.

4 ANOVA: Finding Significant Features

After transforming the target variable and establishing a first statistical understanding of house prices, the next step is to identify which house characteristics have a statistically significant effect on `LogSalePrice`. This is where ANOVA becomes useful.

The goal of this part is not only to test variables mechanically, but to build a first statistical map of what seems to matter in the housing market. We begin with ten candidate features provided in the project instructions. These features describe different aspects of a house: general quality, exterior quality, basement quality, kitchen quality, fireplace quality, central air conditioning, lot shape, land slope, month sold, and year sold.

Table 5: Candidate features tested with one-way ANOVA

#	Feature	Levels	Description
1	`OverallQual`	1–10	Overall material and finish quality
2	`ExterQual`	Po, Fa, TA, Gd, Ex	Exterior material quality
3	`BsmtQual`	None, Po, Fa, TA, Gd, Ex	Basement height quality
4	`KitchenQual`	Po, Fa, TA, Gd, Ex	Kitchen quality
5	`FireplaceQu`	None, Po, Fa, TA, Gd, Ex	Fireplace quality
6	`CentralAir`	N, Y	Central air conditioning
7	`LotShape`	IR3, IR2, IR1, Reg	General shape of property
8	`LandSlope`	Sev, Mod, Gtl	Slope of property
9	`MoSold`	1–12	Month sold
10	`YrSold`	2006–2010	Year sold

4.1 One-Way ANOVA Method

For each feature, a one-way ANOVA model was fitted to test whether the average value of `LogSalePrice` differs across the groups of that feature. The general model can be written as:

$$\texttt{LogSalePrice}_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \tag{8}$$

where μ is the overall mean, α_i is the effect of the i-th group, and ε_{ij} is the residual error.

For each feature, the hypotheses are:

$$H_0 : \text{all group means are equal,} \tag{9}$$
$$H_1 : \text{at least one group mean is different.} \tag{10}$$

A feature is considered statistically significant if its p-value is smaller than the significance level $\alpha = 0.05$.

4.2 One-Way ANOVA Results

The one-way ANOVA results show a clear separation between variables that have a strong statistical relationship with `LogSalePrice` and variables that do not show enough evidence of an effect.

Table 6: One-way ANOVA results for the ten candidate features

Feature	F-statistic	p-value	Significant
OverallQual	332.17	< 0.001	Yes
ExterQual	415.30	6.93×10^{-195}	Yes
KitchenQual	393.32	4.44×10^{-187}	Yes
BsmtQual	364.73	1.48×10^{-175}	Yes
CentralAir	205.67	9.86×10^{-44}	Yes
LotShape	46.73	7.86×10^{-29}	Yes
FireplaceQu	25.59	6.41×10^{-20}	Yes
LandSlope	1.08	0.3388	No
MoSold	0.99	0.4497	No
YrSold	0.74	0.5656	No

The results show that seven variables are statistically significant: `OverallQual`, `ExterQual`, `KitchenQual`, `BsmtQual`, `CentralAir`, `LotShape`, and `FireplaceQu`. These variables have p-values below 0.05, meaning that the average `LogSalePrice` differs significantly across their levels.

The strongest effects are mainly related to quality. This is intuitive: houses with better overall quality, better exterior material, better kitchens, and better basements tend to have higher prices. `CentralAir` is also significant, showing that comfort features can influence value. `LotShape` and `FireplaceQu` are significant as well, although their effects are weaker compared with the main quality variables.

On the other hand, `LandSlope`, `MoSold`, and `YrSold` are not statistically significant in this analysis. This means that, in this dataset, the slope of the land, the month of sale, and the year of sale do not show strong evidence of affecting the average transformed sale price.

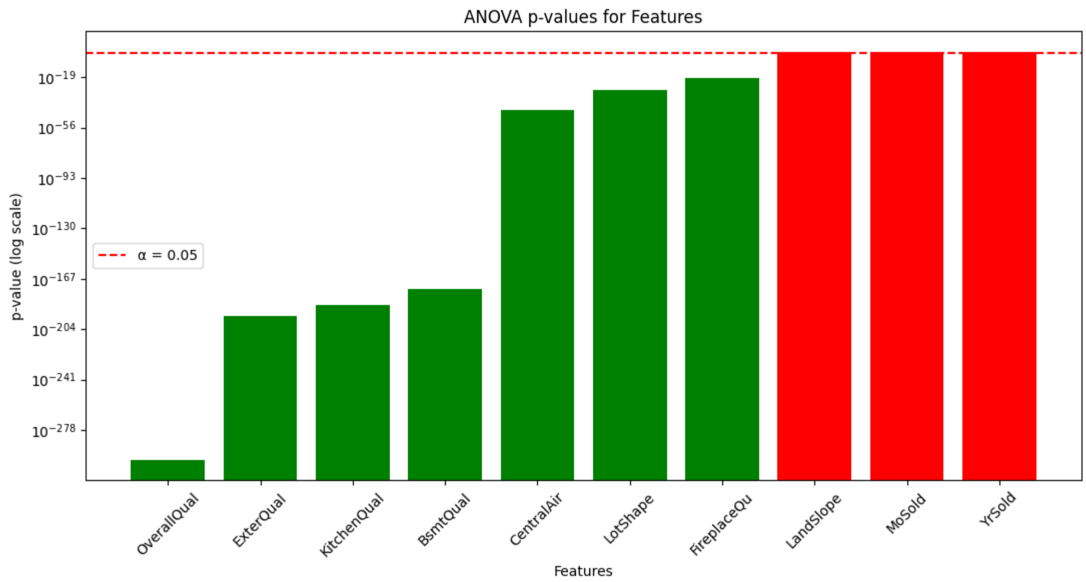

Figure 4: ANOVA p-values for the ten candidate features. Green bars represent significant features, while red bars represent non-significant features.

The p-value plot visually confirms the ANOVA results. Features shown in green have p-values below the threshold $\alpha = 0.05$ and are therefore retained for further analysis. Features shown in red have p-values above the threshold and are not retained.

4.3 Selected Features After One-Way ANOVA

Based on the one-way ANOVA results, the significant features retained for the next steps are:

$$\begin{aligned}&\texttt{OverallQual},\ \texttt{ExterQual},\ \texttt{KitchenQual},\ \texttt{BsmtQual},\\&\texttt{CentralAir},\ \texttt{LotShape},\ \texttt{FireplaceQu}.\end{aligned} \tag{11}$$

The non-significant variables are:

$$\texttt{LandSlope},\ \texttt{MoSold},\ \texttt{YrSold}. \tag{12}$$

This step is important because it reduces the number of variables in a statistically justified way. Instead of moving forward with all candidate variables, the analysis focuses on features that show real evidence of price differences.

4.4 Two-Way ANOVA and Interaction Effects

After identifying the significant individual features, the next question is whether these variables act independently or whether their effects depend on each other. For example, the effect of basement quality may be different depending on the overall quality of the house. To study this, two-way ANOVA models were fitted for all pairs of the seven significant features.

For each pair of variables A and B, the model is:

$$\texttt{LogSalePrice} = \mu + A + B + A \times B + \varepsilon. \tag{13}$$

The term $A \times B$ represents the interaction effect. If this interaction is significant, it means that the effect of one feature on price depends on the level of the other feature.

In total, 21 pairs were tested. Among them, 17 interaction effects were statistically significant. This shows that house prices are not determined only by isolated characteristics, but also by combinations of characteristics.

Table 7: Summary of two-way ANOVA interaction results

Interaction	Significant
`OverallQual` × `KitchenQual`	Yes
`OverallQual` × `BsmtQual`	Yes
`OverallQual` × `CentralAir`	Yes
`OverallQual` × `FireplaceQu`	Yes
`OverallQual` × `LotShape`	Yes
`ExterQual` × `KitchenQual`	Yes
`ExterQual` × `BsmtQual`	Yes
`ExterQual` × `CentralAir`	Yes
`ExterQual` × `FireplaceQu`	Yes
`ExterQual` × `LotShape`	Yes
`KitchenQual` × `BsmtQual`	Yes
`KitchenQual` × `FireplaceQu`	Yes
`KitchenQual` × `LotShape`	Yes
`BsmtQual` × `CentralAir`	Yes
`BsmtQual` × `FireplaceQu`	Yes
`BsmtQual` × `LotShape`	Yes
`CentralAir` × `FireplaceQu`	Yes
`OverallQual` × `ExterQual`	No
`KitchenQual` × `CentralAir`	No
`CentralAir` × `LotShape`	No
`FireplaceQu` × `LotShape`	No

The two-way ANOVA results show that 17 out of 21 tested interactions are statistically significant. This means that most selected features do not act completely independently.

In particular, quality-related variables tend to interact strongly with each other. For example, the value added by a good kitchen or basement may depend on the overall quality level of the house.

However, four interactions are not significant: OverallQual × ExterQual, KitchenQual × CentralAir, CentralAir × LotShape, and FireplaceQu × LotShape. For these pairs, the variables appear to influence house prices more independently.

4.5 Final Two-Way ANOVA Model

The final two-way ANOVA model keeps the seven significant main effects and the 17 significant interaction effects:

$$\begin{aligned}\texttt{LogSalePrice} = \mu &+ \texttt{OverallQual} + \texttt{ExterQual} + \texttt{KitchenQual} + \texttt{BsmtQual} \\ &+ \texttt{CentralAir} + \texttt{FireplaceQu} + \texttt{LotShape} \\ &+ (\texttt{OverallQual} \times \texttt{KitchenQual}) + (\texttt{OverallQual} \times \texttt{BsmtQual}) \\ &+ (\texttt{OverallQual} \times \texttt{CentralAir}) + (\texttt{OverallQual} \times \texttt{FireplaceQu}) \\ &+ (\texttt{OverallQual} \times \texttt{LotShape}) \\ &+ (\texttt{ExterQual} \times \texttt{KitchenQual}) + (\texttt{ExterQual} \times \texttt{BsmtQual}) \\ &+ (\texttt{ExterQual} \times \texttt{CentralAir}) + (\texttt{ExterQual} \times \texttt{FireplaceQu}) \\ &+ (\texttt{ExterQual} \times \texttt{LotShape}) \\ &+ (\texttt{KitchenQual} \times \texttt{BsmtQual}) + (\texttt{KitchenQual} \times \texttt{FireplaceQu}) \\ &+ (\texttt{KitchenQual} \times \texttt{LotShape}) \\ &+ (\texttt{BsmtQual} \times \texttt{CentralAir}) + (\texttt{BsmtQual} \times \texttt{FireplaceQu}) \\ &+ (\texttt{BsmtQual} \times \texttt{LotShape}) + (\texttt{CentralAir} \times \texttt{FireplaceQu}) + \varepsilon. \end{aligned} \tag{14}$$

This model summarizes the main result of the ANOVA analysis: house prices are influenced by both individual features and combinations of features. The price of a house is not simply the sum of isolated characteristics; rather, quality features often reinforce or modify each other's effects.

4.6 Summary of Part 2

The ANOVA analysis identified seven significant features: OverallQual, ExterQual, KitchenQual, BsmtQual, CentralAir, LotShape, and FireplaceQu. These variables showed statistically significant differences in average LogSalePrice across their levels.

The non-significant variables were LandSlope, MoSold, and YrSold. These variables were not retained for the next modeling steps because their p-values were greater than

0.05.

The two-way ANOVA analysis showed that 17 out of 21 interactions were significant. This suggests that house prices are shaped not only by isolated characteristics, but also by combinations of quality-related features.

This part provides the statistical foundation for the next steps of the project. The significant variables identified here will be used in the factorial design and in the parametric regression model.

5 2^k Factorial Design

After identifying the significant features with ANOVA, the next step is to study how selected factors behave together in a controlled factorial framework. The purpose of a 2^k factorial design is to transform selected variables into two levels, usually Low and High, and then study all possible combinations of these levels.

In this project, three factors were selected from the significant ANOVA features:

- Factor A: `OverallQual`, representing overall material and finish quality.
- Factor B: `CentralAir`, representing the presence of central air conditioning.
- Factor C: `BsmtQual`, representing basement quality.

These variables were chosen because they are statistically significant and meaningful from a housing-market perspective. `OverallQual` captures the general quality of the house, `CentralAir` represents comfort, and `BsmtQual` represents the quality of an important structural component.

5.1 Definition of the Factor Levels

To construct a 2^3 factorial design, each selected factor was converted into a binary variable:

Table 8: Definition of the binary factors used in the 2^3 factorial design

Factor	Original variable	Low level (0)	High level (1)
A	`OverallQual`	`OverallQual` < 7	`OverallQual` ≥ 7
B	`CentralAir`	No central air	Central air
C	`BsmtQual`	Other basement quality	Good or excellent basement

This binary transformation creates a clean factorial structure with $2^3 = 8$ possible combinations. Each house is assigned to one of these eight groups depending on whether it has low or high overall quality, no central air or central air, and other or good/excellent basement quality.

5.2 Group Means for the 2^3 Design

The first step in the factorial analysis is to compute the mean `LogSalePrice` for each of the eight combinations. The dollar-equivalent mean price was also computed by applying the inverse logarithmic transformation.

Table 9: Group means for the 2^3 factorial design

Combination	A	B	C	Mean LogPrice	Mean Price
A− B− C−	0	0	0	11.4321	92,239
A− B− C+	0	0	1	11.7309	124,361
A− B+ C−	0	1	0	11.7964	132,772
A− B+ C+	0	1	1	11.9833	160,065
A+ B− C−	1	0	0	12.0364	168,785
A+ B− C+	1	0	1	12.0151	165,236
A+ B+ C−	1	1	0	12.1723	193,359
A+ B+ C+	1	1	1	12.4030	243,521

The group means show a clear price progression. The lowest average price appears in the A− B− C− group, corresponding to houses with lower overall quality, no central air, and no good/excellent basement. The highest average price appears in the A+ B+ C+ group, corresponding to houses with high overall quality, central air, and good/excellent basement quality.

This result is intuitive and important. It suggests that the most valuable houses are not defined by one characteristic alone, but by the accumulation of desirable characteristics. A house with better quality, better comfort, and better basement quality tends to command a much higher sale price.

However, some factor combinations have very small sample sizes. For example, the A+ B− C+ group contains only a very small number of observations. Therefore, some group means should be interpreted with caution.

5.3 Factorial ANOVA Model

To formally test the main effects and interaction effects, the following factorial ANOVA model was fitted:

$$\texttt{LogSalePrice} = \mu + A + B + C + AB + AC + BC + ABC + \varepsilon. \tag{15}$$

In this model, A, B, and C are the main effects. The terms AB, AC, and BC represent two-way interactions, while ABC represents the three-way interaction.

Table 10: ANOVA results for the 2^3 factorial design

Source	Sum of Squares	df	F-statistic	p-value
A	37.6122	1	532.50	1.26×10^{-100}
B	10.0739	1	142.62	2.00×10^{-31}
A:B	0.0716	1	1.01	0.3143
C	9.2086	1	130.37	5.62×10^{-29}
A:C	0.0479	1	0.68	0.4105
B:C	0.0180	1	0.25	0.6137
A:B:C	0.1423	1	2.01	0.1560
Residual	102.5604	1452	–	–

The factorial ANOVA results show that all three main effects are highly significant. Factor A, which represents `OverallQual`, has the largest F-statistic. This confirms that overall house quality is the strongest factor among the three selected variables. Factor B, central air conditioning, and Factor C, basement quality, are also statistically significant.

However, none of the interaction effects are statistically significant at the 5% level. The interaction terms A:B, A:C, B:C, and A:B:C all have p-values greater than 0.05. This suggests that, in this simplified 2^3 factorial design, the three factors mainly act independently. In other words, the effect of one factor does not strongly depend on the levels of the other factors.

5.4 Main Effects and Interaction Effects

To better understand the relative strength of each factor, contrast-based factorial effects were computed on the log scale.

Table 11: Estimated factorial effects on the log-price scale

Effect	Estimated effect
B: CentralAir	0.2846
A: OverallQual	0.2801
C: BsmtQual	0.2341
ABC interaction	0.0909
BC interaction	0.0479
AB interaction	-0.0242
AC interaction	0.0061

The largest estimated effects are the three main effects. Central air conditioning has the largest estimated effect, followed closely by overall quality and basement quality. This means that moving from the low level to the high level of these factors is associated with a clear increase in average `LogSalePrice`.

The interaction effects are much smaller in magnitude. This is consistent with the factorial ANOVA table, where none of the interaction p-values were statistically significant. The small interaction effects suggest that the combined effects of these three factors are mostly additive rather than strongly interactive.

5.5 Interaction Plots

Interaction plots were used to visually examine whether the effect of one factor changes depending on the level of another factor.

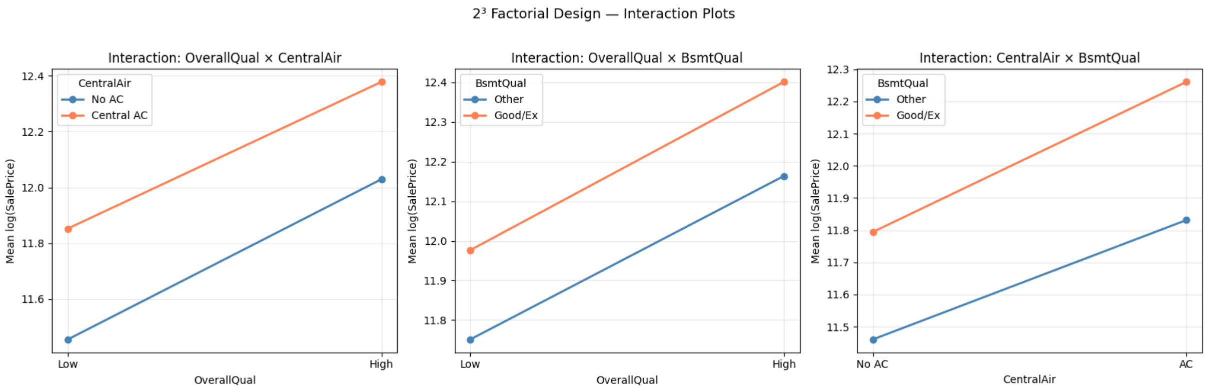

Figure 5: Interaction plots for the 2^3 factorial design. The lines are mostly parallel, suggesting limited interaction effects between the selected factors.

The interaction plots support the ANOVA results. In the three plots, the lines are mostly parallel. Parallel lines indicate that the effect of one factor is similar across the levels of the other factor. This means that the relationship is mainly additive.

For example, houses with good or excellent basement quality tend to have higher average `LogSalePrice` than houses with other basement quality, both for low-quality and high-quality houses. Similarly, houses with central air conditioning tend to have higher prices than houses without central air conditioning. However, the gaps between the lines do not change dramatically, which suggests that the interaction effects are limited.

5.6 Summary of Part 3

The 2^3 factorial design provided a structured way to study three important housing factors together: overall quality, central air conditioning, and basement quality. Each factor was converted into a binary high/low variable, resulting in eight possible factor combinations.

The group means showed a clear price pattern. Houses with low quality, no central air, and weaker basement quality had the lowest average prices, while houses with high quality, central air, and good or excellent basement quality had the highest average prices.

The factorial ANOVA confirmed that all three main effects are statistically significant. However, the two-way and three-way interaction effects were not significant. This suggests that, within this simplified factorial design, house prices are mainly driven by the individual effects of overall quality, central air conditioning, and basement quality, rather than by strong interaction effects between them.

This part complements the previous ANOVA analysis. While the two-way ANOVA in Part 2 showed many significant interactions among the broader set of selected variables, the restricted 2^3 factorial design shows that the three chosen factors behave mostly additively when simplified into high and low levels.

6 Parametric Regression

After the ANOVA and factorial design analyses, the next objective is to build a parametric regression model. The goal of this part is not only to predict `LogSalePrice`, but also to keep the model interpretable. Unlike the final neural network model, which will use all available features, the parametric regression model uses a smaller set of variables selected from the previous statistical analyses.

The regression model is built using the significant features identified by ANOVA, together with two numerical variables required by the project: `GrLivArea` and `TotalBsmtSF`. The target variable is `LogSalePrice`, because the logarithmic transformation reduced the strong skewness of the original `SalePrice` variable and made the modeling task more stable.

6.1 Selected Variables

The selected variables for the parametric regression model are:

- `OverallQual`: overall material and finish quality.
- `ExterQual`: exterior material quality.
- `KitchenQual`: kitchen quality.
- `BsmtQual`: basement quality.
- `CentralAir`: presence of central air conditioning.
- `LotShape`: general shape of the property.
- `FireplaceQu`: fireplace quality.
- `GrLivArea`: above-ground living area.

- `TotalBsmtSF`: total basement area.

These variables represent a balance between statistical significance and practical housing interpretation. Some variables describe quality, others describe comfort, and the numerical variables describe physical size.

6.2 Treatment of Missing Values

Before fitting the regression model, missing values were checked in the selected variables.

Table 12: Missing values before treatment in the regression variables

Variable	Missing values
OverallQual	0
ExterQual	0
KitchenQual	0
BsmtQual	37
CentralAir	0
LotShape	0
FireplaceQu	690
GrLivArea	0
TotalBsmtSF	0
LogSalePrice	0

The missing values in `BsmtQual` and `FireplaceQu` are meaningful. A missing value in `BsmtQual` usually indicates that the house has no basement. Similarly, a missing value in `FireplaceQu` indicates that the house has no fireplace.

Therefore, these observations were not removed. Instead, the missing values were replaced by a new category called `None`. This allows the model to keep the information that the house does not have a basement or fireplace.

6.3 Encoding of Ordinal and Categorical Variables

Linear regression requires numerical input variables. Therefore, categorical and ordinal variables were encoded numerically.

Quality-related variables were mapped from low quality to high quality using the following ordered scale:

Table 13: Encoding of quality-related variables

Category	Encoded value
None	0
Po	1
Fa	2
TA	3
Gd	4
Ex	5

This encoding was applied to `ExterQual`, `KitchenQual`, `BsmtQual`, and `FireplaceQu`. The variable `CentralAir` was encoded as a binary variable, where 1 represents the presence of central air conditioning and 0 represents its absence. The variable `LotShape` was encoded according to regularity, with more regular lots receiving higher values.

Table 14: Additional encodings used in the regression model

Variable	Category	Encoded value
CentralAir	N	0
	Y	1
LotShape	IR3	1
	IR2	2
	IR1	3
	Reg	4

After this encoding step, all selected variables were numerical and could be used in the regression model.

6.4 OLS Regression Model

The first parametric model fitted was an Ordinary Least Squares regression model. The model can be written as:

$$\begin{aligned} \texttt{LogSalePrice} = \beta_0 &+ \beta_1 \texttt{OverallQual} + \beta_2 \texttt{ExterQual} + \beta_3 \texttt{KitchenQual} \\ &+ \beta_4 \texttt{BsmtQual} + \beta_5 \texttt{CentralAir} + \beta_6 \texttt{LotShape} + \beta_7 \texttt{FireplaceQu} \\ &+ \beta_8 \texttt{GrLivArea} + \beta_9 \texttt{TotalBsmtSF} + \varepsilon. \end{aligned} \tag{16}$$

The OLS model was first fitted on the full training dataset in order to interpret the coefficients, their significance, and the global explanatory power of the selected variables.

Table 15: Global summary of the OLS regression model

Quantity	Value
Number of observations	1460
R^2	0.816
Adjusted R^2	0.815
F-statistic	713.8
Prob(F-statistic)	0.000
AIC	-987.7
BIC	-934.9

The model achieves an R^2 value of 0.816 and an adjusted R^2 value of 0.815. This means that approximately 81.6% of the variability in `LogSalePrice` is explained by the selected explanatory variables. The global F-test is highly significant, which confirms that the model provides a statistically meaningful explanation of house prices.

6.5 Interpretation of the Regression Coefficients

The estimated coefficients are presented in Table 16.

Table 16: OLS regression coefficients

Variable	Coefficient	Std. Error	t-statistic	p-value
Intercept	10.4001	0.047	222.933	< 0.001
OverallQual	0.0883	0.006	14.640	< 0.001
ExterQual	0.0491	0.013	3.792	< 0.001
KitchenQual	0.0739	0.010	7.142	< 0.001
BsmtQual	0.0402	0.007	5.506	< 0.001
CentralAir	0.2020	0.019	10.401	< 0.001
LotShape	-0.0412	0.008	-5.111	< 0.001
FireplaceQu	0.0199	0.003	6.670	< 0.001
GrLivArea	0.0002	1.14×10^{-5}	19.397	< 0.001
TotalBsmtSF	9.983×10^{-5}	1.34×10^{-5}	7.424	< 0.001

All explanatory variables have p-values below 0.05. This means that each selected variable is statistically significant in the regression model. This confirms that the variables identified in the ANOVA analysis remain useful when combined in a parametric regression model.

Most coefficients are positive. This means that better quality, larger living area, larger basement area, and the presence of comfort features are generally associated with higher

LogSalePrice. In particular, OverallQual, KitchenQual, CentralAir, GrLivArea, and TotalBsmtSF all contribute positively to house prices.

The coefficient of CentralAir is positive, which suggests that houses equipped with central air conditioning tend to have higher prices after controlling for the other variables.

The coefficient of LotShape is negative in this model. This result should be interpreted carefully. Since LotShape was manually encoded according to regularity, the negative coefficient may be affected by the encoding choice or by relationships with other explanatory variables.

Overall, the OLS model shows strong explanatory power and confirms that house quality, comfort, living area, and basement size are important factors in explaining house prices.

6.6 In-Sample Prediction

After fitting the OLS model, predicted values of LogSalePrice were computed on the training data. The residuals were defined as:

$$e_i = y_i - \hat{y}_i. \tag{17}$$

The in-sample training performance is summarized below.

Table 17: In-sample diagnostic performance of the OLS model

Metric	Value
Training RMSE on LogSalePrice	0.1713
Mean residual	0.0000
Residual standard deviation	0.1714

The mean residual is approximately zero, which is expected in an OLS regression model with an intercept. The training RMSE of 0.1713 gives an in-sample diagnostic of the model's error on the log-price scale. However, this value is computed on the same data used to fit the model, so it should not be interpreted as the final predictive performance.

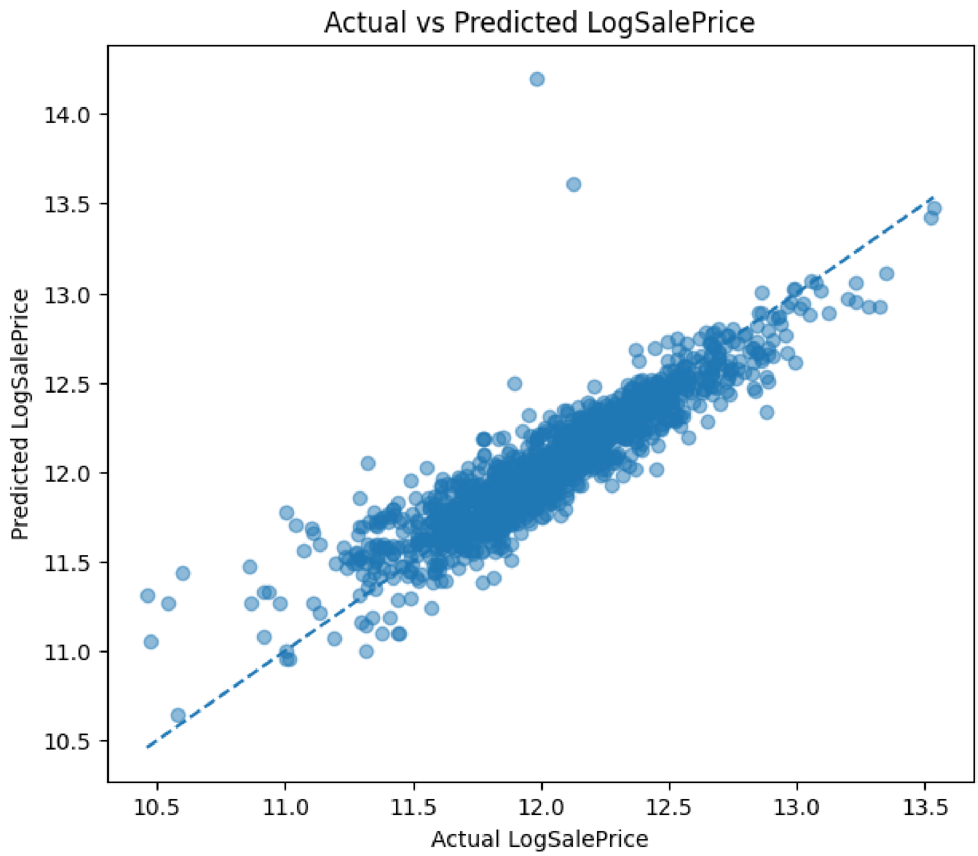

Figure 6: Actual versus predicted `LogSalePrice` for the OLS model. Points close to the diagonal line indicate accurate predictions.

The actual versus predicted plot shows that many points lie close to the diagonal line. This indicates that the OLS model captures the main relationship between the selected explanatory variables and `LogSalePrice`. However, some points deviate from the line, especially near the extremes. This suggests that the linear model explains the main structure of the data but cannot capture all complex or nonlinear patterns.

6.7 Residual Diagnostics

Residual diagnostics were used to evaluate the validity of the OLS model. Ideally, residuals should be centered around zero, show no clear pattern against fitted values, and approximately follow a normal distribution.

Instead of treating each diagnostic plot separately, the three plots are read together in Figure 7. Together, they provide a more complete picture of the model's behavior.

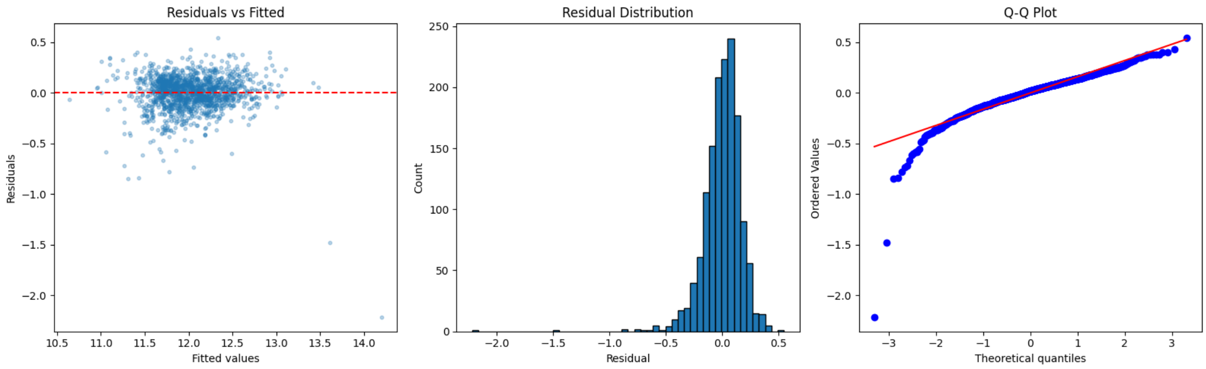

Figure 7: Residual diagnostics for the OLS model: residuals versus fitted values, residual distribution, and Q-Q plot.

The residuals versus fitted plot shows that most residuals are scattered around zero, which suggests that the model does not have a strong systematic bias for the majority of observations. However, a few large negative residuals are visible, especially for high fitted values. These points correspond to houses for which the model overpredicted `LogSalePrice`.

The residual distribution confirms this observation. Most residuals are concentrated near zero, meaning that the model makes relatively small errors for most houses. The central part of the distribution is approximately bell-shaped, but the left tail is longer. This indicates the presence of a few extreme negative errors.

The Q-Q plot gives a similar conclusion. The central residuals are close to the normal reference line, but the lower tail deviates strongly from normality. Therefore, the normality assumption is only approximately satisfied. The deviation is mainly caused by a small number of outlying observations.

Overall, the residual diagnostics show that the OLS model is a reasonable and interpretable benchmark, but not a perfect model. It captures the main linear structure of house prices, while some extreme observations and nonlinear patterns remain difficult to explain.

6.8 Validation Evaluation of Parametric Regression Models

The OLS summary and residual diagnostics are useful for interpretation, but they are computed on the training data. To evaluate predictive performance more realistically, the regression dataset was split into a training subset and a validation subset.

The models were fitted on the training subset and evaluated on the validation subset using RMSE on `LogSalePrice` and validation R^2. This validation evaluation allows us to compare standard Linear Regression, Ridge Regression, and Lasso Regression on unseen observations.

Table 18: Validation performance of the parametric regression models

Model	Validation RMSE	Validation R^2
Linear Regression	0.172853	0.839891
Ridge Regression	0.172873	0.839854
Lasso Regression	0.173128	0.839381

The validation results show that the three parametric regression models perform very similarly. The standard Linear Regression model obtains the lowest validation RMSE, with a value of approximately 0.1729, and a validation R^2 of approximately 0.8399.

Ridge Regression gives almost the same performance, which means that adding an L_2 penalty does not significantly improve the model. Lasso Regression also performs similarly, but its RMSE is slightly higher.

Therefore, regularization does not provide a meaningful improvement for this selected set of predictors. This is reasonable because the regression model uses a limited number of variables selected through ANOVA, so the model is already relatively simple and interpretable.

6.9 Regularized Regression: Ridge and Lasso

Ridge and Lasso regressions were tested as regularized alternatives to the standard linear regression model. These methods add a penalty term to the regression objective in order to improve model stability and reduce overfitting.

Ridge regression uses an L_2 penalty. This penalty shrinks coefficients toward zero, but usually keeps all variables in the model. Lasso regression uses an L_1 penalty. This penalty can shrink some coefficients exactly to zero and can therefore perform variable selection.

In this project, Ridge and Lasso did not improve the validation RMSE compared with the standard Linear Regression model. Therefore, the unregularized linear regression model is kept as the main parametric benchmark.

6.10 ANOVA on the Regression Model

Finally, ANOVA was applied to the fitted regression model to evaluate the contribution of each explanatory variable. This allows us to identify which variables explain the largest part of the variation in `LogSalePrice` within the parametric model.

Table 19: ANOVA table for the parametric regression model

Variable	Sum of Squares	df	F-statistic	p-value
OverallQual	6.3366	1	214.34	2.27×10^{-45}
ExterQual	0.4251	1	14.38	1.56×10^{-4}
KitchenQual	1.5079	1	51.01	1.45×10^{-12}
BsmtQual	0.8961	1	30.31	4.35×10^{-8}
CentralAir	3.1979	1	108.17	1.75×10^{-24}
LotShape	0.7721	1	26.12	3.64×10^{-7}
FireplaceQu	1.3151	1	44.48	3.64×10^{-11}
GrLivArea	11.1232	1	376.26	1.05×10^{-74}
TotalBsmtSF	1.6293	1	55.11	1.93×10^{-13}
Residual	42.8664	1450	–	–

The ANOVA table shows that all variables are statistically significant contributors to the regression model. Among all variables, `GrLivArea` has the largest F-statistic, indicating that above-ground living area is the strongest contributor in the regression model. This is expected, since larger living spaces are usually associated with higher house prices.

`OverallQual` is also a very strong contributor, confirming again the importance of overall construction and finish quality. `CentralAir`, `TotalBsmtSF`, `KitchenQual`, and `FireplaceQu` also make meaningful contributions to explaining `LogSalePrice`.

6.11 Summary of Part 4

The parametric regression model provides an interpretable benchmark before moving to the neural network model. Using selected significant ANOVA features and two important numerical predictors, the OLS model explains approximately 81.6% of the variability in `LogSalePrice`.

The coefficient analysis confirms that house quality, comfort features, living area, and basement size are important determinants of house prices. The residual diagnostics show that the model performs reasonably well, although some deviations from normality and a few large residuals remain. This indicates that the linear model captures the main structure of the data, but it cannot fully model all complex or nonlinear patterns.

The validation comparison shows that Linear Regression, Ridge Regression, and Lasso Regression have very similar predictive performance. Since Ridge and Lasso do not improve the validation RMSE, the standard Linear Regression model is retained as the main parametric benchmark.

7 Non-Parametric Model: Neural Network

The final modeling step is the neural network. Unlike the parametric regression model, which used only a selected group of interpretable variables, the neural network uses all available explanatory features after preprocessing. The objective of this part is to build the model used to generate the final `submission.csv` file for Kaggle evaluation.

The target variable is `LogSalePrice`, because the Kaggle competition evaluates predictions using RMSE on log-transformed prices. The final submitted file, however, must contain predictions on the original `SalePrice` scale. Therefore, predictions are transformed back using the inverse logarithmic transformation before creating the submission file.

Since the Kaggle test set does not contain the true `SalePrice` values, the test RMSE cannot be computed locally. For this reason, a validation set is created from the training data and used to select the best neural network configuration.

7.1 Data Preparation

The first step is to separate the target variable from the explanatory variables. The columns `Id`, `SalePrice`, and `LogSalePrice` are removed from the training features because they should not be used as predictors. The `Id` column from the test set is saved because it is required for the final Kaggle submission file.

The training and test features are then temporarily combined. This ensures that preprocessing steps such as missing value treatment, feature engineering, and categorical encoding are applied consistently to both datasets.

Table 20: Initial feature matrix sizes for the neural network

Dataset	Shape
Training features	(1460, 79)
Test features	(1459, 79)
Combined features	(2919, 79)

This step prepares a single feature matrix containing both training and Kaggle test observations, while keeping the target variable separate.

7.2 Feature Engineering

The original dataset contains many useful variables, but some important housing information is spread across several columns. For example, the total size of a house is not represented by only one variable, but by basement area, first floor area, and second floor area.

To make the learning task easier for the neural network, additional features were created from existing variables. These engineered features summarize important housing characteristics, such as total square footage, total bathrooms, total porch area, house age, remodeling age, and binary indicators for the presence of a basement, garage, fireplace, and pool.

The following engineered variables were added:

- `TotalSF`: total square footage, combining basement, first floor, and second floor areas.
- `TotalBathrooms`: total number of bathrooms, including full and half bathrooms.
- `TotalPorchSF`: total porch area.
- `HouseAge`: age of the house at the time of sale.
- `RemodAge`: time since the last remodeling.
- `HasBasement`, `HasGarage`, `HasFireplace`, and `HasPool`: binary indicators.
- `OverallQual_TotalSF` and `OverallQual_GrLivArea`: interaction features between quality and size.

After feature engineering, the combined dataset increased from 79 to 90 explanatory variables.

Table 21: Feature matrix size after feature engineering

Step	Shape
After feature engineering	(2919, 90)

This step is important because it gives the neural network more informative inputs. Instead of forcing the model to reconstruct complex housing concepts from many separate columns, engineered features directly summarize meaningful property characteristics.

7.3 Log Transformation of Skewed Numerical Features

Some numerical variables have very large ranges and right-skewed distributions. Examples include `LotArea`, `GrLivArea`, `TotalBsmtSF`, and `TotalSF`. These variables may contain a few very large values, which can dominate the scale of the data and make neural network training more difficult.

To reduce the influence of extreme values, selected numerical features were transformed using:

$$x_{\text{new}} = \log(1 + x). \tag{18}$$

The transformation was applied to variables such as LotArea, GrLivArea, TotalBsmtSF, 1stFlrSF, 2ndFlrSF, GarageArea, WoodDeckSF, OpenPorchSF, TotalSF, TotalPorchSF, OverallQual_TotalSF, and OverallQual_GrLivArea.

This transformation compresses large values while preserving the order of the observations. It helps the neural network learn more stable patterns and is consistent with the use of LogSalePrice as the target variable.

7.4 Missing Values and Categorical Encoding

Neural networks require complete numerical input data. Therefore, missing values and categorical variables must be handled before training.

Numerical missing values were replaced by the median of each numerical feature. The median was used because it is robust to extreme values. Categorical missing values were replaced by None, which often represents the absence of a feature in this dataset.

After missing value treatment, categorical variables were converted into numerical dummy variables using one-hot encoding. This creates binary columns indicating whether each observation belongs to a given category.

Table 22: Encoding summary for the neural network

Quantity	Value
Numerical columns	47
Categorical columns	43
Shape after encoding	(2919, 321)
Total missing values after preprocessing	0

After this step, all features were numerical and complete, so the dataset was ready for scaling and model training.

7.5 Train/Validation Split and Feature Scaling

After preprocessing, the combined dataset was split back into training and Kaggle test sets. The training set contains the houses for which SalePrice is known, while the Kaggle test set is kept only for the final submission.

Since the Kaggle test set does not contain the true sale prices, it cannot be used to evaluate RMSE locally. Therefore, the training data was split into a training subset and a validation subset. The validation subset is used to compare neural network configurations and select the best model.

The input features were standardized using `StandardScaler`. This step is important for neural networks because they are trained using gradient-based optimization, and features with very different scales can make training unstable or slower.

Table 23: Train/validation split and scaling summary

Dataset	Shape
Full training feature matrix	(1460, 321)
Kaggle test feature matrix	(1459, 321)
Scaled training subset	(1168, 321)
Scaled validation subset	(292, 321)
Scaled Kaggle test set	(1459, 321)

7.6 Hyperparameter Tuning with Early Stopping

Several neural network configurations were tested using `MLPRegressor`. The goal was to select the architecture and hyperparameters that minimize the validation RMSE on `LogSalePrice`.

The neural network was trained using the Adam optimizer, which is an adaptive version of gradient descent. During training, the model updates its weights step by step to reduce prediction error.

Early stopping was used to reduce overfitting. With early stopping, the optimization process stops when the model no longer improves on an internal validation split. This is important because a neural network can easily fit the training data too closely and lose its ability to generalize to unseen houses.

The target variable was also standardized during training. After prediction, the outputs were transformed back to the original `LogSalePrice` scale before computing RMSE.

Table 24: Neural network hyperparameter tuning results

Architecture	Alpha	Learning rate	Validation RMSE
$(256, 128)$	0.0010	0.0010	0.153672
$(256, 128, 64)$	0.0001	0.0010	0.154795
$(256, 128)$	0.0001	0.0010	0.157257
$(128, 64)$	0.0010	0.0010	0.167627
$(128, 64)$	0.0001	0.0010	0.168048
$(256, 128)$	0.0001	0.0005	0.173320
$(128, 64)$	0.0001	0.0005	0.175056
$(128, 64, 32)$	0.0001	0.0010	0.178989
$(128, 64, 32)$	0.0010	0.0010	0.178998
$(128, 64, 32)$	0.0001	0.0005	0.187162
$(64, 32)$	0.0001	0.0005	0.196120
$(64, 32)$	0.0010	0.0010	0.214417
$(64, 32)$	0.0001	0.0010	0.214424

The best validation performance was obtained with the architecture $(256, 128)$, using `alpha = 0.001` and a learning rate of `0.001`. This model achieved a validation RMSE of approximately 0.1537 and a validation R^2 of approximately 0.8735.

Compared with the first neural network attempt, whose validation RMSE was approximately 0.179, this result is a clear improvement. The improvement comes from three main changes: feature engineering, logarithmic transformations of skewed numerical variables, and systematic hyperparameter tuning.

7.7 Validation Evaluation of the Selected Neural Network

Before generating the Kaggle submission file, the selected neural network configuration was evaluated on the validation set. This is the most meaningful local evaluation because the validation observations were not used to fit the model.

The selected model achieved the following validation performance:

Table 25: Validation performance of the selected neural network

Metric	Value
Validation RMSE	0.1537
Validation R^2	0.8735

The validation RMSE is the main local estimate of generalization performance. It is more meaningful than the training error because it evaluates the model on observations that were not used directly for fitting.

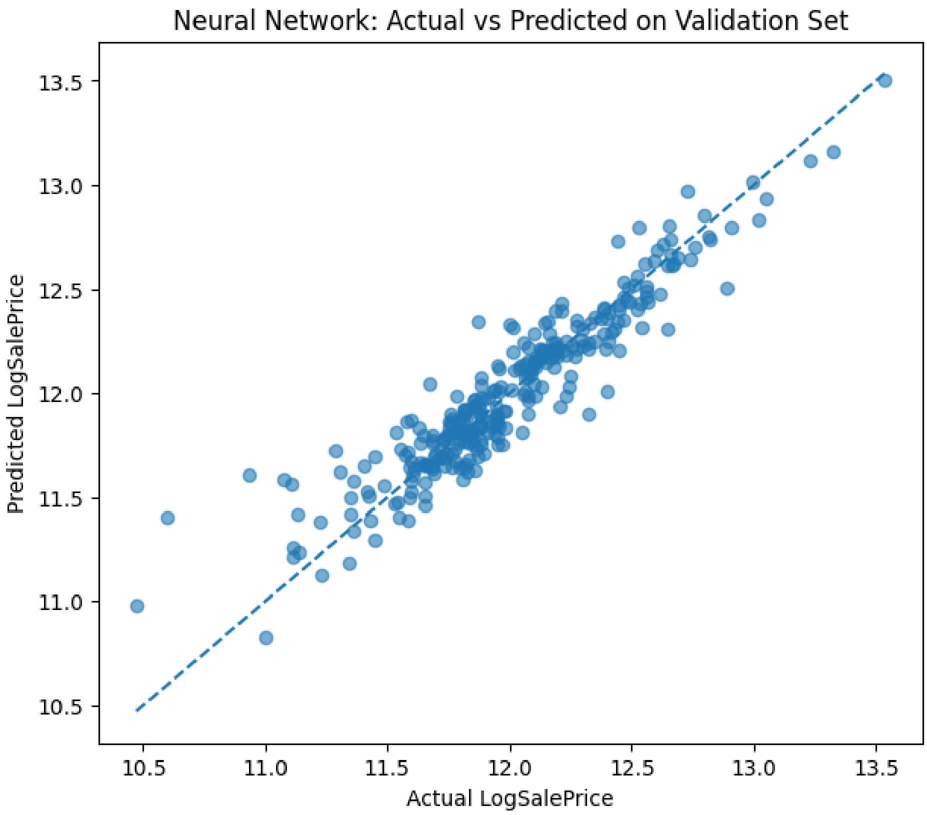

Figure 8: Actual versus predicted `LogSalePrice` values on the validation set for the selected neural network.

The actual versus predicted plot shows that most validation observations lie close to the diagonal line. This indicates that the neural network captures the main relationship between the explanatory variables and `LogSalePrice`. Some dispersion remains, which is expected because the validation set contains unseen observations.

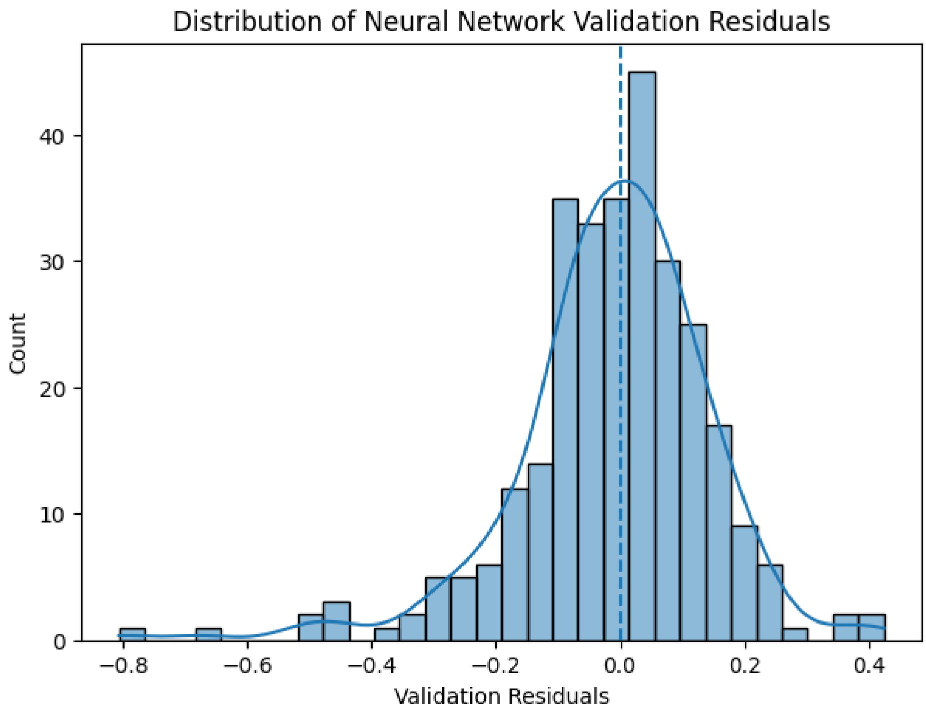

Figure 9: Distribution of validation residuals for the selected neural network.

The validation residuals are centered around zero, which means that the model does not systematically overestimate or underestimate house prices on the validation set. The distribution still contains some larger errors, especially in the tails, but the overall pattern suggests reasonable generalization.

The Kaggle test labels are hidden, so the final RMSE on the test set cannot be computed locally. If the submitted Kaggle score is around 0.19, this means that the hidden test set is more difficult than the local validation split, or that some overfitting remains despite early stopping. This difference is common in prediction competitions and confirms why validation performance should be interpreted as an estimate, not as a guaranteed test score.

7.8 Final Model Retraining and Gradient Descent Visualization

After validating the selected model, the final neural network was retrained using all available training observations. This final model is used to generate predictions for the Kaggle test set.

The final model keeps the best validation configuration:

- Architecture: $(256, 128)$.
- Activation function: ReLU.
- Optimizer: Adam.

- Regularization parameter: $\alpha = 0.001$.
- Learning rate: 0.001.
- Early stopping: activated.

The loss curve in Figure 10 visualizes the gradient-based optimization process. It shows how the training loss changes as the Adam optimizer updates the neural network weights.

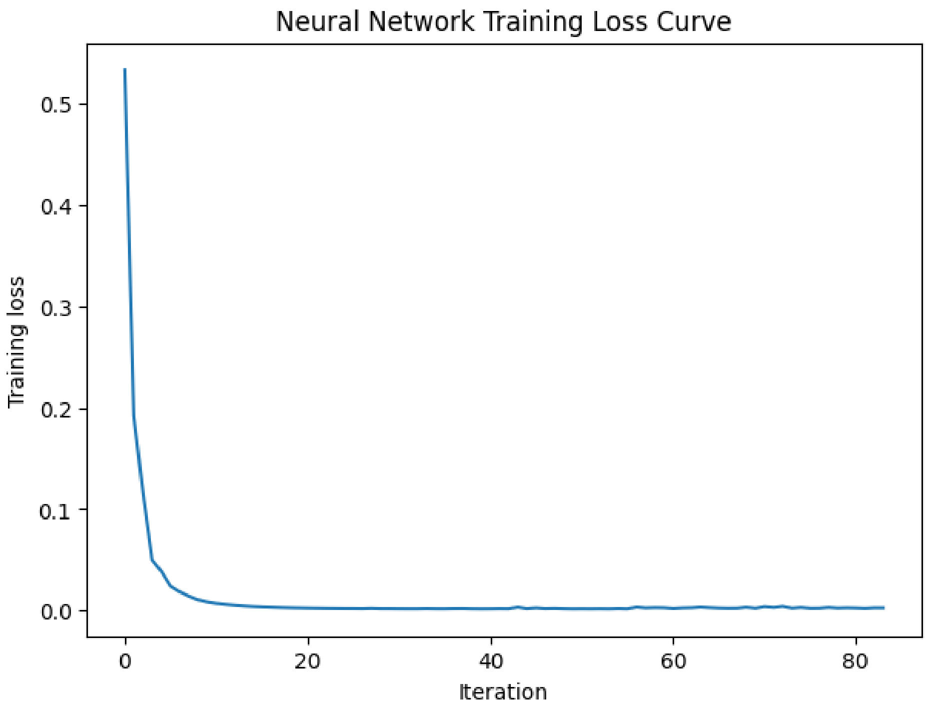

Figure 10: Training loss curve of the final neural network. The sharp decrease at the beginning shows that the Adam optimizer quickly reduces the prediction error.

The loss curve decreases sharply during the first iterations, showing that the Adam optimizer quickly improves the network weights. After several iterations, the curve becomes almost flat, which means that the model has reached a stable solution and additional iterations bring only small improvements.

This confirms that the neural network training is based on an iterative gradient-based optimization process. Since early stopping is activated, training stops when the internal validation performance no longer improves.

7.9 Kaggle Test Prediction and Submission File

The final step is to use the trained neural network to predict house prices for the Kaggle test set. Since the model predicts standardized `LogSalePrice` values, the predictions are first transformed back to the original log-price scale. Then, the inverse logarithmic transformation is applied:

$$\widehat{\texttt{SalePrice}} = \exp(\texttt{Log}\widehat{\texttt{SalePrice}}) - 1. \tag{19}$$

The final submission file contains exactly two columns:

- `Id`: the identifier of each test observation.
- `SalePrice`: the predicted sale price on the original dollar scale.

Table 26: Structure of the generated Kaggle submission file

Quantity	Value
Number of rows	1459
Number of columns	2
Missing values in `Id`	0
Missing values in `SalePrice`	0

The generated file `submission.csv` is therefore correctly formatted for Kaggle evaluation. The official prediction score is computed externally by Kaggle using the hidden test labels and the RMSE on log-transformed prices.

7.10 Summary of Part 5

The neural network model was developed as the final predictive model of the project. It used all available explanatory features after extensive preprocessing, including feature engineering, logarithmic transformations, missing value treatment, one-hot encoding, and feature scaling.

Several neural network configurations were compared using validation RMSE with early stopping. The best model used the architecture $(256, 128)$, with $\alpha = 0.001$ and a learning rate of 0.001. It achieved a validation RMSE of approximately 0.1537 and a validation R^2 of approximately 0.8735.

The validation results show that the neural network improves over the earlier parametric regression benchmark. However, the Kaggle score may be higher than the validation RMSE because it is computed on a hidden test set. This highlights the importance of distinguishing between local validation performance and external test performance.

Finally, the trained neural network was used to generate the required `submission.csv` file. This completes the predictive part of the project and connects the statistical analysis to the Kaggle evaluation framework.

8 Conclusion

This project followed a complete statistical workflow to understand and predict house prices. The analysis started with the transformation of `SalePrice` into `LogSalePrice`, which reduced skewness and made the target variable more suitable for statistical modeling.

Classical inference provided a first understanding of the average log-price and its variability. Then, ANOVA identified the most important housing characteristics, especially quality-related variables such as `OverallQual`, `ExterQual`, `KitchenQual`, and `BsmtQual`. The factorial design showed that overall quality, central air conditioning, and basement quality have significant individual effects on house prices.

The parametric regression model provided an interpretable benchmark. It showed that quality, living area, basement size, and comfort features explain a large part of the variation in `LogSalePrice`. However, residual diagnostics also showed that some nonlinear patterns and extreme observations remain difficult for a linear model to capture.

Finally, a neural network model was trained using all available features after preprocessing and feature engineering. The best model achieved a validation RMSE of approximately 0.1537 and was used to generate the final `submission.csv` file for Kaggle evaluation.

Overall, the project shows that house prices are mainly influenced by quality, size, and comfort-related features. It also shows the value of combining statistical interpretation with machine learning: classical methods help explain the data, while the neural network provides a more flexible predictive model.